Computer Architecture Homework-1

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1) a) addi $x_8, x_5, -5$
Expl: subi does not exist in RISC-V, so we use addi with a negative immediate value to achieve the same result.

b) slli $x_5, x_3, 3$
Expl: Instead of using mul which is a costlier operation, we use left shift by 3 bits which is equivalent to multiplying the number by 8.

c) addi x_{19}, x_{19}, x_{10}
Expl: $x_{19} += x_{10}$ is equivalent to $x_{19} = x_{19} + x_{10}$ and hence we use the add instruction with x_{19} appearing twice so we can assign the value stored in x_{19} as $x_{19} + x_{10}$.

d) addi $x_{15}, x_{15}, 1$
Expl: $x_{15}++$ is equivalent to $x_{15} = x_{15} + 1$ which can be done in assembly language using addi.

e) srli $x_9, x_{15}, 2$
Expl: divi does not exist in the RISC-V ISA and hence we use srli to shift the bits right by 2 which is equivalent to dividing the number by 4.

f) addi $x_{19}, x_{19}, 24$
Expl: Adding a variable value in a register to a fixed value (immediate) requires us to use addi.

2) a) ld $x_6, 160(x_5)$
addi $x_6, x_6, 100$
sd $x_6, 96(x_5)$

b) ld $x_6, 160(x_5)$
addi $x_6, x_6, 1$
sd $x_6, 160(x_5)$

c) ld x_6, 40(x_5)
ld x_7, 96(x_5)
sd x_7, 40(x_5)
sd x_6, 96(x_5)

d) ld x_6, 32(x_5)
slli x_6, x_6, 32
srli x_7, x_7, 32
sd, x_7, 32(x_5)

e) ld x_6, 16(x_5)
slli x_7, x_6, 32
srli x_8, x_6, 32
or x_9, x_7, x_8
sd x_9, 16(x_5)

3)a) $+23 = 11 \times 2 + 1$
$11 = 5 \times 2 + 1$
$5 = 2 \times 2 + 1$
$2 = 1 \times 2 + 0$
$\therefore +23 = 0000\ 0111$

b) -1: 1111 1111
$+1 = 0000\ 0001$
$-1 = 1111\ 1110 + 0000\ 0001$
$-1 = 1111\ 1111$

c) $+255$: Cannot be represented in 2's complement form for 8 bits as the range of integers that can be represented in this form for 8 bits is -2^8 to $2^8 - 1$ or -128 to 127 which does not include 255.

d) -128:
$+128 = 2^7$
$+128$: 1000 0000 (Unsigned form)
-128: 0111 1111 + 0000 0001
-128: 1000 0000
↳ lowest value representable

4) a) 1101 0100 has MSB=1 and therefore represents a negative value in 2's complement representation.
Complement of 1101 0100 = 0010 1011 = $2^0+2^1+2^3+2^5 = 43$
represents a positive integer.
∴ 1+1101 0100+0010 1011 = 1 0000 0000 = 0 in 2's complement in 8 bits
∴ 1101 0100+1 = -0010 1011
∴ 1101 0100 = -44

b) 0010 1011 represents a +ve integer
As we have seen in part a, it represents 43.
∴ 0010 1011 = 43

c) 1111 1110 has MSB=1 → -ve integer
Complement of 1111 1110 is 0000 0001
0000 0001+0000 0001 = 0000 0010 = 2
∴ 1111 1110 = -2